

Chapter 23

Lecture 23: Compactifications. The Stone-Čech compactification (existence).

Definition 23.0.1. A compactification of a Hausdorff space X is a compact Hausdorff space $Y \supseteq X$ such that $Y = \overline{X}$. We say that compactifications Y_1 and Y_2 are equivalent if there exists a homeomorphism $h : Y_1 \rightarrow Y_2$ such that $h|_X = \text{id}_X$.

Example 23.0.1. $X = (0, 1)$, $Y = [0, 1] \subseteq \mathbb{R}$.

Theorem 23.0.1 (Imbedding theorem). Let X be a T_1 space. Suppose that $\{f_\alpha\}_{\alpha \in J}$ is an indexed family of continuous functions $f_\alpha : X \rightarrow \mathbb{R}$ satisfying the requirement that for each point x_0 of X and each open set U containing x_0 , there exists an index α such that $f_\alpha(x_0)$ is positive and f_α vanishes on $X \setminus U$. Then the function $F : X \rightarrow \mathbb{R}^J$ defined by

$$F(x) = (f_\alpha(x))_{\alpha \in J} \quad (23.1)$$

is an imbedding of X in $\mathbb{R}^J$. If $f_\alpha : X \rightarrow [0, 1]$ for each $\alpha \in J$, then F is an imbedding of X in the cube $[0, 1]^J$.

Proof The proof is similar to the proof of Urysohn metrization theorem.

First, F is continuous because $\mathbb{R}^J$ has the product topology and each f_α is continuous. Second, since X is T_1 , singletons are closed, so for any two distinct points $x, y \in X$, there exists an index α such that $f_\alpha(x) \neq f_\alpha(y)$. So $F(x) \neq F(y)$, which implies that F is injective.

Finally, we must prove that F is a homeomorphism onto its image, the subspace $Z = F(X)$ of $\mathbb{R}^J$. We know that F defines a continuous bijection from X to Z . So it suffices to show that for each open set U in X , $F(U)$ is open in Z . Let z_0 be a point in $F(U)$. We shall find an open set W of Z such that

$$z_0 \in W \subseteq F(U). \quad (23.2)$$

Let x_0 be the point in U such that $F(x_0) = z_0$. Choose an index β for which $f_\beta(x_0)$ is positive and $f_\beta(X \setminus U) = \{0\}$. Let

$$V = \{z \in Z : z_\beta > 0\} = \pi_\beta^{-1}((0, \infty)) \cap Z, \quad (23.3)$$

which is open in Z by the definition of subspace topology. We assert that $z_0 \in V \subseteq F(U)$. First, $z_0 \in V$ because

$$\pi_\beta(z_0) = f_\beta(x_0) > 0. \quad (23.4)$$

Second, $V \subseteq F(U)$. For if $z \in V$, then $\pi_\beta(z) > 0$, so $f_\beta(x) > 0$ for some $x \in U$. Since f_β vanishes on $X \setminus U$, we have $x \in U$. So $z = F(x)$ for some $x \in U$, which means $z \in F(U)$.

Remark Here T_1 is required to guarantee that singletons are closed, so that if $x \neq y$, there exists f_α such that $f_\alpha(x) \neq f_\alpha(y)$. ■

Theorem 23.0.2. A space is completely regular if and only if it is homeomorphic to a subspace of a cube $[0, 1]^J$ for some index set J .

Proof

($\Rightarrow$) Let X be completely regular. For each $x_0 \in X$ and each open set U containing x_0 , there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x_0) = 1$ and f vanishes on $X \setminus U$. Let $\{f_\alpha\}_{\alpha \in J}$ be the collection

of all such continuous functions. Then by the previous theorem, the function $F : X \rightarrow [0, 1]^J$ defined by $F(x) = (f_\alpha(x))_{\alpha \in J}$ is an embedding. Thus X is homeomorphic to the subspace $F(X)$ of the cube $[0, 1]^J$.

($\Leftarrow$) Recall that every metrizable space is normal thus is completely regular. And recall that product of completely regular spaces is completely regular. Since $[0, 1]$ is metrizable, it is completely regular. Thus the cube $[0, 1]^J$ is completely regular.

Since X is homeomorphic to a subspace of $[0, 1]^J$, X is completely regular as well because subspace of completely regular space is completely regular. ■

Lemma 23.0.1. *Let X be a Hausdorff space. Let $h : X \rightarrow Z$ be an imbedding of X in the compact Hausdorff space Z . Then there exists a corresponding compactification Y of X such that there is an imbedding $H : Y \rightarrow Z$ with $H|_X = h$. Such a compactification Y is uniquely determined up to equivalence.*

Proof

Given h , let X_0 denote the subspace $h(X)$ of Z , and let Y_0 denote its closure in Z . Then Y_0 is a compact Hausdorff space and $\overline{X_0} = Y_0$; therefore, Y_0 is a compactification of X_0 .

We now construct a space Y containing X such that the pair (X, Y) is homeomorphic to the pair (X_0, Y_0) . Let us choose a set A disjoint from X that is in bijective correspondence with $Y_0 \setminus X_0$ under some map $k : A \rightarrow Y_0 \setminus X_0$. Let $Y = X \cup A$ and define a bijective correspondence $H : Y \rightarrow Y_0$ by $H(x) = h(x)$ for $x \in X$ and $H(a) = k(a)$ for $a \in A$. Then topologize Y by declaring U to be open in Y if and only if $H(U)$ is open in Y_0 . The map $H : Y \rightarrow Y_0$ is automatically a homeomorphism; and the space X is a subspace of Y because H equals the homeomorphism h when restricted to the subspace X of Y . By expanding the range of H , we obtain the required imbedding of Y into Z .

Now suppose Y_i is a compactification of X and that $H_i : Y_i \rightarrow Z$ is an imbedding that is an extension of h , for $i = 1, 2$. Now H_i maps X onto $h(X) = X_0$. Because H_i is continuous, it must map Y_i into $\overline{X_0}$ (Recall H is continuous iff $H(\overline{X}) \subseteq \overline{H(X)}$); because $H_i(Y_i)$ contains X_0 and is closed (being compact), it contains $\overline{X_0}$. Hence $H_i(Y_i) = \overline{X_0}$, and $H_2^{-1} \circ H_1$ defines a homeomorphism of Y_1 with Y_2 that equals the identity on X . ■

Example 23.0.2. $X = (0, 1)$, $Y = S^1 \subseteq \mathbb{R}^2$, $X \ni x \mapsto (\cos 2\pi x, \sin 2\pi x) \in Y$ is a compactification of X . (Not the Usual sense of Compactification, see the above lemma.)

Property 23.0.1. *Let $X \subseteq Y$ and Y be a compact Hausdorff space. Then X is completely regular.*

Proof

Since Y is compact Hausdorff, Y is normal. Then Y is completely regular. Thus X is completely regular as well. ■

Claim If X is completely regular, it has a compactification.

Proof Since X is completely regular, by Theorem 23.0.2, there exists an embedding $h : X \rightarrow [0, 1]^J$ for some index set J . Since $[0, 1]^J$ is compact Hausdorff, by Lemma 23.0.1, there exists a compactification Y of X . ■

Example 23.0.3. Let $X = (0, 1)$. Consider the embedding $h : X \hookrightarrow \mathbb{R}^2$ defined by $h(x) = (x, \sin \frac{1}{x})$. Let $A = \{0\} \times [-1, 1] \cup \{(1, \sin(1))\}$. Then $Y = h(X) \cup A$ is a compactification of $h(X)$.

Let Y be a compactification of X in $\mathbb{R}^2$ as above. Then the continuous function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = \sin \frac{1}{x}$ can be extended to a continuous function $\bar{f} : Y \rightarrow \mathbb{R}$ by defining $\bar{f} = \pi_2 \circ H$ where $H : Y \rightarrow \mathbb{R}^2$ is the imbedding and $\pi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the projection to the second coordinate.

Theorem 23.0.3 (Stone-Čech compactification). *Let X be a completely regular space. Then there exists a compactification Y of X such that every bounded continuous function $f : X \rightarrow \mathbb{R}$ extends uniquely to a continuous function $\bar{f} : Y \rightarrow \mathbb{R}$. (Y is called the Stone-Čech compactification of X)*

Proof

We prove the existence first and the uniqueness will be discussed in the next lecture.

Proof

Let $\{f_\alpha\}_{\alpha \in J}$ be the collection of all bounded continuous real-valued functions on X , indexed by some set J . For each $\alpha \in J$, choose a closed interval I_α in $\mathbb{R}$ that contains the range of f_α . To be definite, choose

$$I_\alpha = [\inf f_\alpha(X), \sup f_\alpha(X)]. \quad (23.5)$$

Then define $h : X \rightarrow \prod_{\alpha \in J} I_\alpha$ by $h(x) = (f_\alpha(x))_{\alpha \in J}$.

By the Tychonoff theorem, $\prod_{\alpha \in J} I_\alpha$ is compact. Because X is completely regular, points and closed sets can be separated by continuous functions from the family $\{f_\alpha\}_{\alpha \in J}$. So by the Imbedding theorem 23.0.1, h is an imbedding of X in the compact Hausdorff space $\prod_{\alpha \in J} I_\alpha$.

Let Y be the compactification of X induced by h as in Lemma 23.0.1. Then by the lemma there is an imbedding $H : Y \rightarrow \prod_{\alpha \in J} I_\alpha$ that equals h when restricted to the subspace X of Y .

Given a bounded continuous function $f : X \rightarrow \mathbb{R}$, we show it extends to Y . The function belongs to the collection $\{f_\alpha\}_{\alpha \in J}$, so it equals f_β for some $\beta \in J$. Let $\pi_\beta : \prod_{\alpha \in J} I_\alpha \rightarrow I_\beta$ be the projection to the β -th coordinate. Then the continuous map $\bar{f} = \pi_\beta \circ H : Y \rightarrow I_\beta$ is an extension of f for if $x \in X$, then

$$\bar{f}(x) = \pi_\beta(H(x)) = \pi_\beta(h(x)) = f_\beta(x) = f(x). \quad (23.6)$$

■